

Interval POMDP Shielding for Imperfect-Perception Agents

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Abstract—Autonomous systems that rely on learned perception can make unsafe decisions when sensor readings are misclassified. We study shielding for this setting: given a proposed action, a shield blocks actions that could violate safety. We consider the common case where system dynamics are known but perception uncertainty must be estimated from finite labeled data. From these data we build confidence intervals for the probabilities of perception outcomes and use them to model the system as a finite Interval Partially Observable Markov Decision Process with discrete states and actions. We then propose an algorithm to compute a conservative set of beliefs over the underlying state that is consistent with the observations seen so far. This enables us to construct a runtime shield that comes with a finite-horizon guarantee: with high probability over the training data, if the true perception uncertainty rates lie within the learned intervals, then every action admitted by the shield satisfies a stated lower bound on safety. Experiments on four case studies show that our shielding approach (and variants derived from it) improves the safety of the system over state-of-the-art baselines.

I. INTRODUCTION

The computational capabilities unlocked by neural networks have made it feasible to build autonomous agents that use learned components to interact with their environments while pursuing complex goals. As one example, consider an autonomous aircraft taxiing system that senses the environment through a camera and must follow the runway centerline in TaxiNet-style settings [1], [2], [3]. The agent *senses* the environment through sensors, *perceives* the underlying state of the system from those sensor readings via neural components, *chooses* an action based on the perceived state using symbolic or neural decision logic, and *executes* that action to update the system state. When the perception component misclassifies the underlying state from the sensor readings, the agent acts on incorrect information. We refer to this phenomenon as *imperfect-perception*.

Autonomous agents with imperfect-perception are often deployed in safety-critical settings, so one would like all executed actions to come with formal safety guarantees. Shielding is a promising approach: a shield restricts the actions available to the agent so that safety constraints are preserved at runtime [4], [5], [6]. Shields are also compatible with reinforcement-learning (RL)-based control, where they can improve the runtime safety of a deployed RL policy while preserving as much task performance as possible [4], [5], [6].

Constructing a shield with a meaningful safety guarantee requires an accurate model of perception uncertainty. Models of perception uncertainty take many forms, but at their heart they capture the probabilistic relationship between observations received by the controller and the true state of the system. When these observations are produced using a Deep Neural Network (DNN) component the perception uncertainty model acts as a probabilistic black box around the environment producing inputs for the DNN perceptor and the DNN perceptor itself. When constructing this probabilistic model, classical uncertainty models in control theory are often a poor

fit because neural perception uncertainty need not follow simple parametric forms such as Gaussian noise [7]. The support-based shield of Carr *et al.*, which discards emission probabilities and only retains perception relationships, is sound [8], but we show it may be too conservative to yield usable shields in problems of interest.

One central challenge to constructing a tenable probabilistic uncertainty model is the fact that the perception environment experienced by an agent will never exactly match the data from which the uncertainty model was built. This change manifests as a distributional shift in model probabilities. A model’s design determines how robust it is to different kinds of distributional shift. One type of shift results from a change in the control policy from when the data was collected to when the agent is deployed. This changes the distribution of states the agent visits [9], but it does not change the probability associated with producing an observation from a particular state, called an *emission probability*. There are several uncertainty modeling techniques which are not robust to *control-based shift*. One is conformal wrappers, which modify the neural perceptor to be set estimators with a guaranteed lower bound on the probability of the true outcome being contained within the set [10], [11] and have been used for shielding [12]. This guarantee relies on exchangeability, a property that can be violated by control-based shift without additional correction [13], [14]. To guard against control-based shift one can isolate the individual per-state emission probabilities because these probabilities are independent of the true state distribution experienced by the agent. However, this method is not robust to a more pernicious type of distributional shift, *deployment shift*, which changes the per-state emission probabilities themselves. Deployment shift is inherent to our problem setting because the perception uncertainty model is learned from finite data, guaranteeing the sampled probabilities will not exactly match their deployment realizations [15]. Our goal is to build a model robust to control-based shift that guards against deployment shift to the extent that is feasible.

We begin with a Partially Observable Markov Decision Process (POMDP) model of the agent (or system) with a discrete state and action spaces, which is an appropriate base model in this setting as it describes a control system with limited access to state data [16]. In this paper we work throughout with finite discrete models: discrete latent states, discrete actions, and a finite set of discrete observations output by the perception stack. Specifically, a POMDP composes a *dynamics model* of how the state space evolves, with a *perception uncertainty model* of per-state emission probabilities [17], [18]. We address a common regime where the system dynamics are modeled exactly, but the perception uncertainty model is built from finite labeled data. Our method constructs confidence bounds on the emission probabilities and uses them to define an Interval Partially Observable Markov decision process (IPOMDP), whose perception uncertainty model uses intervals for these emission

probabilities rather than fixed values.

This follows the broader robust/imprecise probabilistic modeling perspective of bounded-parameter MDPs and imprecise hidden Markov models [19], [20], [21]. These intervals make the model robust to deployment shift because they anticipate the realized emission probabilities varying from scalar expectations. Like POMDPs, the perception uncertainty model in the IPOMDP is independent of the control policy used to collect the data for estimating emission probabilities/intervals, but unlike the base POMDP model, it also carries a Probably Approximately Correct (PAC)-style model-correctness guarantee: with probability at least λ , the true perception uncertainty model lies in the admissible set of perception uncertainty models induced by the interval bounds [22].

Given this interval perception model, we develop a shielding method that starts from a perfect-perception shield, where the agent is assumed to have access to the true state of the system, and lifts it to the imperfect-perception setting. The underlying perfect-perception shield may come from Probabilistically Controlled Invariant Sets (PCIS) [23], barrier-certificate methods [24], or other state-space shield-synthesis procedures. Starting from such a shield is deliberate: computing a shield directly in belief space under interval observations scales poorly because the belief update is set-valued and normalized, producing high-dimensional nonconvex reachable sets [25]. Our method therefore propagates over-approximations of reachable beliefs and tests action admissibility to the shield through worst-case linear objectives over that envelope, solved with linear programming. The resulting design is modular by allowing various perfect-perception shield designs, conservative by construction, and separates dynamics-level safety shielding from reasoning about perception uncertainty.

We evaluate the method on four benchmark domains: TaxiNet, Obstacle, CartPole, and Refuel. The experiments address two main empirical questions. First, across these benchmarks, how do different shields compare in safety and usability under imperfect-perception? To answer this, we compare our proposed shield with two under-approximate variants of our shield construction and two other baselines including the support-based shield of Carr *et al.* [8]. Second, how conservative is the envelope abstraction used by the proposed method? To answer this, we compare the envelope's over-approximation of reachable beliefs against forward-sampled under-approximations. Together, these experiments show that interval belief-envelope shielding is most useful in partially observable regimes where shields constructed assuming point emission probabilities are too optimistic and the support-based shield becomes too coarse, while also revealing benchmark-dependent computational limits. **These results show a scalability tradeoff: the sound over-approximate shield construction is tractable on smaller IPOMDPs but does not scale to the largest models, while a sampling-based under-approximate variant inspired by the same construction scales further but is not given the same soundness guarantee here.**

Contributions

We make the following contributions:

- 1) An IPOMDP formulation of the imperfect-perception system under study, including finite-data observation uncertainty and the associated shielding problem for finite discrete models.
- 2) A runtime shield construction for IPOMDPs that lifts a perfect-perception shield to the imperfect-perception setting.

- 3) **A sound but expensive linear-programming/linear-fractional-programming-based method for over-approximating reachable beliefs in an IPOMDP, as well as a cheap, under-approximate, sampling-based variant, and a method for using those belief envelopes for conservative action admissibility.**
- 4) Empirical evidence across four benchmark domains (TaxiNet, Obstacle, CartPole, and Refuel), including a comparison among Observation, Single-Belief, Envelope, Fwd-Sampling, and the support-based shield, and a quantitative study of envelope conservativeness.
- 5) A probabilistic-automata/categorical interpretation of the construction that explains the convex abstraction used for interval belief reasoning; available in supplementary material.

II. PRELIMINARIES

A. System Model

We begin with a discrete Markov Decision Process (MDP) model of the agent/system,

$$\mathcal{M} = (S, A, T),$$

where S is a finite state set, A is a finite action set, and $T: S \times A \times S \rightarrow [0, 1]$ is the transition kernel. At time t , the system is in state $s_t \in S$, the controller chooses action $a_t \in A$, and the next state is sampled from $T(s_t, a_t, \cdot)$. **For a fixed action a , we write T_a for the $|S| \times |S|$ transition matrix with entries $(T_a)_{s, s'} = T(s, a, s')$.**

Perfect perception: Under perfect perception, the controller observes s_t exactly before selecting a_t . In that setting the decision problem is an MDP over the true state space S .

Imperfect perception and raw sensor inputs: In learned autonomy, the controller typically does not observe s_t directly. Instead, a perception component receives a raw sensor input $y_t \in \mathcal{Y}$, where $\mathcal{Y}$ may be high-dimensional, for example an image space. Although y_t is the direct result of s_t , the relationship which produces y_t involves environmental conditions and therefore is extremely difficult to model directly. A learned perception component, such as a deep neural network, then converts y_t into a finite symbolic output. This output is what the controller *observes* in the system model, and forms the basis of the belief of state space occupation.

B. System Model with Probabilistic Abstraction

Let O be a finite set of discrete observations. Rather than model the raw sensor domain $\mathcal{Y}$ directly, we apply previous work [17], [18] to summarize the perception pipeline by a conditional distribution from the true state (s_t) of the system at time t to discrete perception outputs,

$$Z(o|s) = \Pr[\text{perception outputs } o | s_t = s], \quad o \in O.$$

The type $Z: S \times O \rightarrow [0, 1]$ fits a POMDP-style model of the imperfect-perception system: the true state evolves according to T , remains hidden from the controller, and the controller instead receives observations drawn from $Z(\cdot | s)$. The abstraction avoids carrying the full high-dimensional sensor domain through the system model: all sensor complexity is compressed into the finite perception uncertainty model Z , which records only the probabilities of the perception outputs relevant to decision making. Throughout the paper we stay in this finite discrete setting, so

the shield reasons over discrete latent states, discrete actions, and discrete observation symbols rather than continuous sensor values directly. This probabilistic abstraction is close in spirit to recent approaches that also replace the raw sensor domain by a compact stochastic interface between true state and downstream reasoning. Schäfers *et al.* use a learned perception uncertainty model to produce probabilistic state summaries for POMDP belief updates [18], while Păsăreanu *et al.* use confusion-matrix-derived probabilistic abstractions to enable closed-loop probabilistic analysis of vision-based autonomy [17]. Our work shares this abstraction viewpoint, but then develops an interval-valued version of Z to support conservative shielding under finite-data uncertainty.

POMDP: With a fixed (perception uncertainty model) Z , the imperfect-perception system is a POMDP

$$\mathcal{P} = (S, A, O, T, Z).$$

We write $\Delta(S)$ for the probability simplex over S . For a history h_t of action, observation pairs, a belief $b_t \in \Delta(S)$ is the posterior distribution over the true state given that history, i.e.,

$$b_t(s) = \Pr[s_t = s | h_t].$$

Given prior belief $b_t \in \Delta(S)$, action a_t , and received observation o_{t+1} , the controller updates its belief by Bayesian filtering [16]. The state still evolves according to T ; the uncertainty comes from not observing s_t directly and instead receiving only o_t through the perception uncertainty model Z .

C. Interval POMDP System Model

In practice we must estimate the true Z from finite data. While one can form point estimates of the emission probabilities, finite-sample error can make those estimates arbitrarily misleading, especially for rarely observed state-observation pairs. We therefore replace the single perception uncertainty model by interval bounds $Z^-, Z^+ : S \times O \rightarrow [0, 1]$ [17], [26] and define the admissible set of perception uncertainty models

$$\mathcal{Z} = \left\{ Z \mid \begin{array}{l} \forall s, o: Z^-(o|s) \leq Z(o|s) \leq Z^+(o|s), \\ \forall s: \sum_o Z(o|s) = 1. \end{array} \right\}.$$

Definition II.1 (IPOMDP). An IPOMDP is a tuple

$$\mathcal{I} = (S, A, O, T, Z^-, Z^+),$$

where (S, A, T) is the initial MDP system and $\mathcal{Z}$ is the admissible set of perception uncertainty models defined above.

Given history $h_t = (a_0, o_1, \dots, a_{t-1}, o_t)$ and prior belief $b_0 \in \Delta(S)$, each $Z \in \mathcal{Z}$ induces a posterior belief b_t^Z . Over all elements of $\mathcal{Z}$, this forms a set of beliefs:

$$\mathcal{B}_t(b_0, h_t) = \{b_t^Z : Z \in \mathcal{Z}\}.$$

This definition fixes a single unknown Z across the whole history h_t . For shielding we use a more conservative semantics that allows the perception uncertainty model to vary within $\mathcal{Z}$ at each step; as a result our belief-envelope construction is a superset of the belief set produced by the fixed semantics. We adopt this stronger per-step-varying semantics because the safety guarantee should remain valid for any admissible realization of the perception error at

each step, rather than depending on one fixed but unknown kernel being sampled once and then held constant for the whole trajectory. There is also a computational reason for this: the deployed method propagates a compact belief envelope at runtime, which scales much better than recomputing the exact shield from the entire action-observation history at each step.

In an ordinary POMDP, a single posterior belief is updated by Bayesian filtering. In an IPOMDP, however, interval uncertainty in the perception uncertainty model means that a whole set of posterior beliefs can be consistent with the same history, so we will need a separate algorithm to propagate a conservative belief envelope over a history.

PAC guarantee: When the interval bounds are constructed from finite labeled data using confidence intervals, the resulting IPOMDP carries a dataset-level model-correctness guarantee. In particular, if the true perception uncertainty model is denoted by Z^* , then our construction ensures

$$\Pr[Z^* \in \mathcal{Z}] \geq \lambda,$$

for a user-chosen confidence level λ . Section IV gives the concrete Clopper–Pearson construction and the corresponding union-bound argument.

D. Shielding

Our objective is to lift a state-based (perfect-perception) safety shield to the imperfect-perception setting rather than synthesize an entirely new shield.

Definition II.2 (Perfect-Perception Shield). A perfect-perception shield is a map $\Omega : S \rightarrow 2^A$ assigning each true state the set of admissible actions.

The shield is intended to enforce a safety specification defined by a designated safe set of states specified logically. It removes actions whose probability of leaving that safe set exceeds the chosen guarantee notion. In other words, we consider safety specifications that can be expressed using the safety fragment of Probabilistic Computational Tree Logic (PCTL).

This definition provides a minimal interface – a map from true states to a set of actions – and is agnostic to how the shield is synthesized. It may come from reachability analysis, barrier-certificate methods, controlled invariant sets, or any other state-space shield construction.

Definition II.3 (Imperfect-Perception Shield). An imperfect-perception shield is a map $\Xi : \bigcup_{t \geq 0} \mathcal{H}_t \rightarrow 2^A$ from finite action-observation histories to sets of actions.

Here $\mathcal{H}_t$ denotes the set of length- t action-observation histories, e.g. $h_t = (a_0, o_1, \dots, a_{t-1}, o_t) \in (A \times O)^t$. This is the trajectory-level shielding object: given the history available at time t , it returns the actions permitted at that time.

A runtime shield, by contrast, is the runtime mechanism used to realize such a map. It maintains an internal summary of state uncertainty, updates that summary after each action-observation pair, and computes the next permitted-action set. In this way, the runtime shield induces an imperfect-perception shield over histories. This distinction matters because although the shield itself is fundamentally history-based, our implementation is designed to avoid storing and recomputing over the full history explicitly.

To define the runtime shield, first define the action-safety indicator from the perfect-perception shield

$$\chi_{\Omega}(s,a) = \begin{cases} 1, & a \in \Omega(s), \\ 0, & a \notin \Omega(s). \end{cases}$$

For each action a , let

$$\phi_a(b) = \sum_{s \in S} b(s) \chi_{\Omega}(s,a),$$

so $\phi_a(b)$ is the probability, under belief b , that action a is allowed by the perfect-perception shield.

For a history h_t , initial belief b_0 , and threshold $\beta \in [0,1]$, the exact lifted shield induced by Ω is

$$\Xi^*(h_t) = \left\{ a \in A : \inf_{b \in \mathcal{B}_t(b_0, h_t)} \phi_a(b) \geq \beta \right\}. \quad (1)$$

That is, an action is permitted only if it remains Ω -admissible with probability at least β under every belief in the interval-POMDP reachable-belief set $\mathcal{B}_t(b_0, h_t)$. Our method approximates this exact object conservatively using belief envelopes that over-approximate that reachable-belief set.

PCIS construction: We use Probabilistically Controlled Invariant Sets (PCIS) as the main source of perfect-perception shields because they provide a composable stochastic invariance certificate [23]. In that framework, an N -step ϵ -PCIS is a set of states C such that each state in C admits a policy that keeps the trajectory inside C for N steps with probability at least ϵ ; the infinite-horizon variant requires this for all future times.

To obtain a perfect-perception shield from a certified invariant set $C \subseteq S$, we define

$$\Omega(s) = \left\{ a \in A : \sum_{s' \in C} T(s, a, s') \geq \gamma \right\}, \quad (2)$$

with threshold $\gamma \in (0,1]$. Thus, PCIS supplies the invariant core C , and Ω is the induced state-action shield under the known dynamics kernel T . The remainder of the paper only requires the interface $\Omega: S \rightarrow 2^A$, so the runtime lifting method is compatible with other perfect-perception shield constructions as well.

III. METHOD

Because Ω is synthesized offline in true-state space, the runtime shield maintains a conservative belief set consistent with the action-observation history and admits only actions safe for every belief in that set. We make this tractable by reducing belief-envelope propagation to a linear optimization problem over a template domain, yielding an LP-solvable over-approximation.

Our goal is to propagate not just one belief but an entire belief envelope through an action-observation update, so we begin with the algorithm to propagate a single belief, the ordinary POMDP forward recursion, equivalently the Bayes-filter update [27]. Given a prior belief b and action-observation pair (a,o) , the unnormalized posterior mass is

$$u_{s'} = Z(o|s') \sum_{s \in S} T(s, a, s') b(s),$$

and the posterior belief is obtained by normalizing u . Equivalently, in vector form,

$$y = T_a^\top b, \quad u_s = y_s Z(o|s), \quad b'_s = \frac{u_s}{\sum_j u_j}.$$

Our envelope update follows exactly this recursion, but simultaneously for every prior belief in $\hat{\mathcal{B}}_t$ and every perception uncertainty model in $\mathcal{Z}$. The LP conservatively applies this recursion over these sets to construct an outer approximation of the reachable beliefs.

A. Exact Admissibility Objective

Using the belief-level safety score ϕ_a from Section II, given history h_t , exact admissibility at threshold β is

$$a \in \Xi^*(h_t) \iff \inf_{b \in \mathcal{B}_t(b_0, h_t)} \phi_a(b) \geq \beta.$$

This is exactly the shield defined in Equation (1), written using ϕ_a for compactness. Since $\mathcal{B}_t$ is generally intractable to represent exactly, we replace it with the conservative envelope $\hat{\mathcal{B}}_t$ in the deployed runtime shield below.

B. Why Exact Convex-Hull Propagation is Intractable

The exact lifted shield depends on the reachable-belief set $\mathcal{B}_t$, so the main algorithmic question is how to represent and propagate that set online. A natural first idea is to propagate the exact reachable set itself, but that approach is intractable.

The dynamical state update step of the IPOMDP that applies the transition kernel T is affine in the belief, so it preserves convexity by itself. The difficulty appears in the observation update: the posterior is obtained by multiplying by emission probability intervals and then normalizing by the total evidence mass, which introduces bilinear and linear-fractional terms. As a result, even when the current belief set is convex, the exact set of reachable posteriors can become nonconvex after one update. Intuitively, an extreme point of this reachable set corresponds to a worst-case combination of two ingredients: a corner of the current belief set and a particular choice of endpoint probabilities within the observation intervals. After one step, each such corner case can branch into many new corner cases, because a different endpoint choice can be worst-case for each state and observation. Repeating this process over time causes the number of relevant extreme points to grow exponentially with the horizon. This vertex explosion makes exact convex-hull propagation scale very poorly in both memory and runtime, which is why it is not suitable for online shielding [25].

C. Outer Polytope Propagation

We describe how to propagate the belief envelope through the IPOMDP for a single step, i.e., a single application of the dynamics and observation update under a fixed action a . We represent each belief envelope by a *template polytope*: a polytope described by a fixed set of linear directions chosen in advance, together with the worst-case bound attained in each direction. Equivalently, the template matrix specifies which halfspaces we track, and the vector of offsets records how far the reachable set extends along those directions. At time t , maintain

$$\hat{\mathcal{B}}_t = \{b \in \mathbb{R}^{|S|} : A_t b \leq d_t, b \geq 0, \mathbf{1}^\top b = 1\}, \quad \mathcal{B}_t \subseteq \hat{\mathcal{B}}_t.$$

Let $n = |S|$. If the current envelope has m_t template rows, then $A_t \in \mathbb{R}^{m_t \times n}$ and $d_t \in \mathbb{R}^{m_t}$.

A key structural reason for using convex envelopes is that the dynamical state update step is affine in the current belief, so sets of reachable pre-normalization beliefs remain convex; the loss of convexity enters only after the observation update, where the uncertain products $y_s w_s$ are coupled with the Bayes normalization by the total evidence mass. Our goal is not to propagate one posterior belief, but an entire prior belief envelope through an action-observation update. The difficulty is that the exact update does not fit directly into LP: uncertain observation probabilities create bilinear products, and Bayes normalization introduces division by the total evidence mass. The purpose of the next steps is therefore to rewrite the propagation problem into a conservative LP-solvable form.

Steps 1–3 describe the update for one candidate prior belief $b \in \tilde{B}_t$ and give the corresponding LP construction immediately after the intuition. Step 4 then assembles the final belief envelope by optimizing template directions over all prior beliefs and all admissible observation-probability choices. We write each LP in the standard form

$$\max_x r^\top x \quad \text{s.t.} \quad Gx \leq h, Ex = f, \underline{x} \leq x \leq \bar{x}.$$

Here $Gx \leq h$ collects inequality constraints, $Ex = f$ collects equality constraints, $\underline{x} \leq x \leq \bar{x}$ gives simple bounds, and r is the objective vector; the complete LP is the tuple $(r, E, f, G, h, \underline{x}, \bar{x})$.

Step 1 (Dynamical state update): Assuming that the agent chooses action a , we can propagate a prior belief through the known dynamics as,

$$y = T_a^\top b.$$

Here T_a is the transition matrix for action a . This step computes the next-state belief based on the known dynamics function. The vector y is the belief after the action a but before conditioning on the observation. This map is affine in b , so convexity is preserved and the update remains LP-compatible. In the LP, this step contributes the prior-normalization and state-update constraints $\mathbf{1}^\top b = 1$ and $y = T_a^\top b$. Because T_a is row-stochastic, meaning its entries are nonnegative and each row sums to one, these constraints preserve total mass: $b \geq 0$ and $\mathbf{1}^\top b = 1$ make the prior a distribution, and y is also a distribution.

Step 2 (Observation update without normalization): At this point the current observation o arrives. Assuming that w denotes the observation probabilities, the updated unnormalized belief is given by,

$$Z^-(o|s) \leq w_s \leq Z^+(o|s), \quad u_s = y_s w_s \quad \forall s.$$

For each state s , the scalar w_s denotes the observation probability $Z(o|s)$ chosen from the admissible interval for that state-observation pair. The quantity u_s is the resulting unnormalized posterior mass contributed by state s : post-dynamical update belief for state s , y_s , weighted by the probability that state s produces observation o . In an ordinary POMDP w_s would be fixed; in an IPOMDP w_s may vary within the interval, so we must account for all admissible choices. The scalar $\sum_j u_j$ is the total evidence mass used in the normalization step below. So u_s is intentionally state-wise, while $\sum_j u_j$ is the quantity obtained after summing those state-wise contributions across all s .

The coupling term is bilinear, so we replace $u_s = y_s w_s$ by its McCormick envelope [28] over known bounds on y_s and w_s . This gives a sound linear over-approximation of feasible (u, y, w) tuples. Intuitively, the McCormick envelope is the standard LP relaxation of a bounded product: it replaces the exact bilinear equality by linear inequalities that still contain every feasible product value. In this sense, the resulting optimization is an LP relaxation of the forward recursion over the whole belief envelope rather than a different update rule. The construction introduces the observation-probability vector w and unnormalized mass vector u , and collects the pre-normalization variables into the LP decision vector in a fixed block order, meaning the variable groups are concatenated in the same order used by the LP constraint matrices:

$$z = (b^\top, y^\top, w^\top, u^\top)^\top \in \mathbb{R}^{4n}.$$

With this block order, the equality constraints from Step 1 are

$$E_0 = \begin{bmatrix} \mathbf{1}^\top & 0 & 0 & 0 \\ -T_a^\top & I & 0 & 0 \end{bmatrix}, \quad f_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

so $E_0 z = f_0$ enforces $\mathbf{1}^\top b = 1$ and $y = T_a^\top b$. The prior envelope itself enters as the first block row of the inequality matrix below. The construction then restricts the interval observation row to the observed output o while respecting row-stochasticity:

$$w_s^- = \max \{Z^-(o|s), 1 - \sum_{o' \neq o} Z^+(o'|s)\},$$

$$w_s^+ = \min \{Z^+(o|s), 1 - \sum_{o' \neq o} Z^-(o'|s)\}.$$

For each state, compute state-update bounds

$$L_s = \min_{b \in \tilde{B}_t} e_s^\top T_a^\top b, \quad U_s = \max_{b \in \tilde{B}_t} e_s^\top T_a^\top b,$$

where e_s is the standard basis vector. These bounds define the McCormick relaxation [28] of $u_s = y_s w_s$ over $y_s \in [L_s, U_s]$ and $w_s \in [w_s^-, w_s^+]$. With block columns ordered as (b, y, w, u) , writing D_v for a diagonal matrix and $\odot$ for componentwise multiplication, the inequality data for these unnormalized constraints are

$$G_0 = \begin{bmatrix} A_t & 0 & 0 & 0 \\ 0 & D_{w^-} & D_L & -I \\ 0 & D_{w^+} & D_U & -I \\ 0 & -D_{w^-} & -D_U & I \\ 0 & -D_{w^+} & -D_L & I \end{bmatrix}, \quad h_0 = \begin{bmatrix} d_t \\ L \odot w^- \\ U \odot w^+ \\ -U \odot w^- \\ -L \odot w^+ \end{bmatrix}.$$

The first block row is the prior-envelope constraint $A_t b \leq d_t$; the last four block rows are the McCormick inequalities. Together with $E_0 z = f_0$, the simple bounds are

$$\underline{z} = (0^\top, L^\top, (w^-)^\top, 0^\top)^\top, \quad \bar{z} = (1^\top, U^\top, (w^+)^\top, 1^\top)^\top.$$

Thus $G_0 z \leq h_0$, $E_0 z = f_0$, and the simple bounds define the unnormalized feasible set.

Step 3 (Normalization): Posterior belief is fractional,

$$b'_s = \frac{u_s}{\sum_j u_j}.$$

This is the usual Bayes normalization: divide each unnormalized posterior mass by the total evidence mass so the posterior sums to one. That division is exactly what breaks linearity. To recover LP solvability, we apply Charnes–Cooper [29]: introduce a

scaling variable for $(\sum_j u_j)^{-1}$ and scale decision variables so normalization constraints become linear.

A useful way to view this step is to work first with unnormalized posterior masses, where the update is affine, and then recover normalized beliefs through a linear-fractional normalization handled by Charnes–Cooper [29]. Intuitively, this rewrites the posterior update so that division by the total evidence mass is absorbed into the new variables, allowing normalized posterior queries to be solved by LP without changing their optimal values. For a template direction c , the normalized objective before transformation is the linear-fractional program

$$\max \frac{c^\top u}{\mathbf{1}^\top u} \quad \text{s.t.} \quad G_0 z \leq h_0, E_0 z = f_0, z \leq \bar{z}.$$

Choose an arbitrary scale constant $\kappa > 0$, let $\eta = \mathbf{1}^\top u$, and define $\tilde{z} = (\kappa/\eta)z$ and $t = \kappa/\eta$. With $x = (\tilde{z}^\top, t)^\top$, $d_u \in \mathbb{R}^{4n}$ selecting the u -block, and block order $(\tilde{b}, \tilde{y}, \tilde{w}, \tilde{u}, t)$, the standard LP tuple is

$$r = (0^\top, 0^\top, 0^\top, c^\top, 0)^\top,$$

$$G = \begin{bmatrix} G_0 & -h_0 \\ I & -\bar{z} \\ -I & \underline{z} \end{bmatrix}, \quad h = 0, \quad E = \begin{bmatrix} E_0 & -f_0 \\ d_u^\top & 0 \end{bmatrix}, \quad f = \begin{bmatrix} 0 \\ \kappa \end{bmatrix},$$

with $\underline{x} = (-\infty, \dots, -\infty, 0)^\top$ and $\bar{x} = (\infty, \dots, \infty, \infty)^\top$. The inequality matrix G is the scaled copy of $G_0 z \leq h_0$ together with the scaled variable bounds $\underline{z} \leq z \leq \bar{z}$. The equality $d_u^\top \tilde{z} = \kappa$ fixes the normalization scale. The LP objective equals $\kappa c^\top u / (\mathbf{1}^\top u)$, so dividing the optimum by κ gives the actual template offset.

Step 4 (Template projection): Lift the single-belief update to the whole envelope. Fix the next template matrix $A_{t+1} \in \mathbb{R}^{m_{t+1} \times n}$, with one row per next-envelope template direction, and define

$$\hat{B}_{t+1} = \{b' \in \mathbb{R}^{|S|} : A_{t+1} b' \leq d_{t+1}, b' \geq 0, \mathbf{1}^\top b' = 1\},$$

where each component of the offset vector d_{t+1} is computed by one LP (with $c_i^\top$ the i th row of A_{t+1}):

$$d_{t+1,i} = \sup_{c_i^\top} b',$$

In this optimization, $A_t, d_t, A_{t+1}, a, o, Z^-, Z^+$ are fixed inputs. The LP variables are the prior belief b , predicted belief y , admissible observation-probability vector w , unnormalized mass u , and the auxiliary Charnes–Cooper variables that encode the normalized posterior. Lower template bounds are obtained by using direction $-c_i$ and negating the resulting optimum. Step 4 is where the whole prior envelope is propagated: each LP ranges over every prior belief in $\hat{B}_t$ together with every admissible choice for the observation probabilities from the intervals and returns the worst-case extent of the reachable posterior set in one template direction. After solving this LP for every row $c_i^\top$ of A_{t+1} , the offsets $d_{t+1,i}$ assemble the next envelope.

Facet-wise maximization gives the tightest bound along each chosen template direction while keeping runtime propagation tractable. Geometrically, each row of the template matrix asks how far the reachable posterior set extends in one chosen direction in belief space; for example, with coordinate directions, the template polytope simply records upper and lower bounds on each state’s belief mass.

Facet-wise maximization costs one LP per template direction. If $n = |S|$ and m_t is the number of constraints in $A_t b \leq d_t$, a typical lifted LP has $O(n)$ variables (roughly $4n$ to $5n$) and $O(m_t + n)$ constraints, plus four McCormick inequalities per state. With k_{t+1}

template rows, propagation solves k_{t+1} LPs per action-observation update, so runtime scales mainly with template facets and action-observation branching; Fig. 5 evaluates this empirically.

The shape of $\hat{B}_{t+1}$ is a template polytope in H -representation: an intersection of halfspaces with the probability simplex. This choice makes worst-case linear safety queries and propagation share the same LP machinery. In this work we use hypercube templates, i.e., coordinate-wise bounds on belief mass. Other template families, such as zonotopes, ellipsoids, or vertex-based polytopes, could be used instead, with the usual tradeoff that richer templates can tighten the envelope at higher computational cost.

D. Conservative Shield

Define lower bound

$$\underline{p}_t(a) = \inf_{b \in \hat{B}_t} \phi_a(b).$$

Since $\phi_a(b) = \sum_s b_s \chi_\Omega(s, a)$, this score is the ordinary LP

$$\underline{p}_t(a) = \min_b \sum_s b_s \chi_\Omega(s, a) \quad \text{s.t.} \quad A_t b \leq d_t, b \geq 0, \mathbf{1}^\top b = 1,$$

with no Charnes–Cooper variables. The deployed shield is

$$\hat{\Xi}(h_t) = \{a \in A : \underline{p}_t(a) \geq \beta\}.$$

The replacement of B_t with $\hat{B}_t$ in the infimum calculation is conservative because $\hat{B}_t$ contains every belief in B_t , so minimizing over the larger set can only decrease the safety score and therefore can only remove actions, not admit unsound ones.

E. Two-State Worked Example

As a numerical check, take an IPOMDP with two states $s = 1, 2$, identity dynamics, and a current belief envelope $\hat{B} = \{b : 0.4 \leq b_1 \leq 0.6, b_2 = 1 - b_1\}$, where b_i is the belief mass on state i . Let the observation alphabet be $\{o, \bar{o}\}$, and suppose the observed output at this step is o . Suppose the row interval constraints are

	$Z(o s)$	$Z(\bar{o} s)$
$s = 1$	$[0.65, 0.95]$	$[0.10, 0.30]$
$s = 2$	$[0.10, 0.50]$	$[0.60, 0.80]$

with row-stochastic observation rows. Projecting onto the received symbol gives $w_1 \in [0.7, 0.9]$ and $w_2 \in [0.2, 0.4]$. Then

$$b'_1 = \frac{b_1 w_1}{b_1 w_1 + (1 - b_1) w_2}.$$

Using coordinate template directions $c = e_1$ and $c = -e_1$, this expression is increasing in b_1 and w_1 , and decreasing in w_2 . Thus the lower endpoint, equivalently the solution of the LP in direction $-e_1$, occurs at $b_1 = 0.4, w_1 = 0.7, w_2 = 0.4$:

$$(b'_1)^{\min} = \frac{0.4 \cdot 0.7}{0.4 \cdot 0.7 + 0.6 \cdot 0.4} = \frac{7}{13} \approx 0.538.$$

With $\kappa = 1$, this Charnes–Cooper solution has evidence $\eta = 0.52$, scale $t = 25/13$, and posterior $b' = (7/13, 6/13)$. The upper endpoint occurs at $b_1 = 0.6, w_1 = 0.9, w_2 = 0.2$, where the row-stochastic complementary probabilities are $Z(\bar{o}|1) = 0.10$ and $Z(\bar{o}|2) = 0.80$:

$$(b'_1)^{\max} = \frac{0.6 \cdot 0.9}{0.6 \cdot 0.9 + 0.4 \cdot 0.2} = \frac{27}{31} \approx 0.871.$$

The corresponding Charnes–Cooper solution has evidence $\eta = 0.62$, scale $t = 50/31$, and posterior $b' = (27/31, 4/31)$. Thus the propagated coordinate envelope is $0.538 \leq b'_1 \leq 0.871$, $b'_2 = 1 - b'_1$. If action a is safe only in state 1, then $\phi_a(b') = b'_1$, the conservative score is $7/13$, and the lifted shield admits a exactly for thresholds $\beta \leq 7/13$, blocking it for thresholds such as $\beta = 0.6$.

IV. INTERVAL CONSTRUCTION AND PAC CORRECTNESS

We now show how the interval bounds are constructed from finite labeled data and why the resulting admissible set of perception uncertainty models contains the true perception uncertainty model with high probability. This section supplies the statistical side of the method: the previous sections defined how shielding uses the admissible set of perception uncertainty models online, and here we explain how those interval bounds are learned offline with a finite-sample correctness guarantee.

A. Clopper–Pearson Construction

For each true state s and observation o , let n_s be the number of samples with true label s , and $k_{s,o}$ the count observed as o . We use these counts to estimate the emission probability $Z(o|s)$ and to construct a confidence interval for that probability. For confidence level $1 - \alpha_{s,o}$, the corresponding Clopper–Pearson bounds are

$$Z^-(o|s) = \text{BetaInv}\left(\frac{\alpha_{s,o}}{2}; k_{s,o}, n_s - k_{s,o} + 1\right), \quad (3)$$

$$Z^+(o|s) = \text{BetaInv}\left(1 - \frac{\alpha_{s,o}}{2}; k_{s,o} + 1, n_s - k_{s,o}\right), \quad (4)$$

where $\text{BetaInv}(q; a, b)$ denotes the q -quantile (inverse CDF) of a $\text{Beta}(a, b)$ distribution. These are the exact equal-tail binomial confidence bounds for the entry $Z(o|s)$. For the edge cases, we use the standard Clopper–Pearson conventions: if $k_{s,o} = 0$, then $Z^-(o|s) = 0$ and $Z^+(o|s) = \text{BetaInv}(1 - \frac{\alpha_{s,o}}{2}; 1, n_s)$; if $k_{s,o} = n_s$, then $Z^-(o|s) = \text{BetaInv}(\frac{\alpha_{s,o}}{2}; n_s, 1)$ and $Z^+(o|s) = 1$. These interval endpoints define the admissible set of perception uncertainty models $\mathcal{Z}$ together with the simplex constraints $\sum_o Z(o|s) = 1$ for each state s . In the sequel we work directly with the interval bounds $Z^-(o|s)$ and $Z^+(o|s)$ rather than constructing $\mathcal{Z}$ explicitly.

We use Clopper–Pearson rather than Wilson intervals because Theorem V.2 later needs a finite-sample guarantee that, with probability at least λ over the random training dataset, the admissible set of perception uncertainty models contains the true perception uncertainty model. To obtain that dataset-level containment statement, we first need valid entry-wise coverage for each (s, o) before applying a union bound. Clopper–Pearson provides exact binomial coverage for every sample size and parameter value, whereas Wilson intervals can undercover in finite samples, especially near the boundary [22], [30], [31]. We therefore prefer Clopper–Pearson because guaranteed coverage is needed for the later PAC-style containment result, even at the cost of wider intervals. The practical consequence is that the learned interval bounds, and hence the deployed shield, become more conservative: we trade some permissiveness for a stronger finite-sample correctness guarantee.

B. From Intervals to Outer Probability λ

This subsection converts the entry-wise confidence intervals above into the dataset-level correctness probability used throughout

the paper. The relevant quantity is the outer probability from Section II: probability over the random labeled training dataset used to construct the interval bounds. Our goal is to show that, with probability at least λ , the admissible set of perception uncertainty models contains the true perception uncertainty model Z^* entry-wise. Let

$$\mathcal{E}_{\text{corr}} = \bigcap_{s \in S, o \in O} Z^*(o|s) \in [Z^-(o|s), Z^+(o|s)]$$

denote the event that every entry of the true perception uncertainty model lies inside its corresponding learned confidence interval. By union bound, if the entry-wise failure probabilities satisfy

$$\sum_{s \in S} \sum_{o \in O} \alpha_{s,o} \leq \alpha,$$

where $\alpha_{s,o}$ is the probability that the confidence interval for entry (s, o) fails to contain the true value $Z^*(o|s)$, then

$$\Pr(\mathcal{E}_{\text{corr}}) \geq 1 - \alpha \triangleq \lambda.$$

In words, λ is the probability, over the random training dataset used to construct the intervals, that the admissible set of perception uncertainty models $\mathcal{Z}$ contains the true perception uncertainty model Z^* . This is the dataset-level correctness probability that appears in Theorem V.2.

Why the union bound is conservative: The union bound makes no structural assumptions about the observation process: it holds even when the entry-wise correctness events $\{Z^*(o|s) \in [Z^-(o|s), Z^+(o|s)]\}$ are arbitrarily dependent. In practice, these events are typically correlated. For a fixed true state s , all counts $\{k_{s,o}\}_{o \in O}$ are computed from the same pool of n_s samples, so increasing the empirical mass assigned to one observation necessarily reduces the mass available to others. Because of this coupling across observations within the same state, summing the per-entry failure probabilities can substantially under-estimate $\Pr(\mathcal{E}_{\text{corr}})$.

If additional assumptions are defensible, this outer bound can be tightened. For example, under independence of the per-entry failure events one has

$$\Pr(\mathcal{E}_{\text{corr}}) \geq \prod_{s \in S} \prod_{o \in O} (1 - \alpha_{s,o}),$$

which allows selecting larger per-entry $\alpha_{s,o}$ for the same target λ than the simple constraint $\sum_{s,o} \alpha_{s,o} \leq \alpha$. More generally, any joint (simultaneous) confidence region construction that exploits structure—such as row-wise multinomial constraints, state-wise independence, or parametric assumptions on the perception uncertainty model—can replace the Bonferroni/union-bound.

V. GUARANTEES

This section gives two guarantees. The first is a conditional abstraction-soundness statement: if the true perception uncertainty model lies in the admissible set of perception uncertainty models, then any action admitted by the deployed shield is guaranteed to be Ω -admissible with probability at least β . This follows from the construction of the belief envelope, which maintains $\mathcal{B}_t \subseteq \hat{\mathcal{B}}_t$. The second adds the dataset-level correctness probability from Section IV: when Ω is instantiated from a PCIS-style safe subset $C \subseteq S$, we combine abstraction soundness with the probability λ that the admissible set of perception uncertainty models contains the true perception uncertainty model to obtain a finite-horizon safety guarantee.

Theorem V.1 (Abstraction Soundness). *Assume $Z^* \in \mathcal{Z}$. Then for any $a \in \hat{\Xi}(h_t)$ and the true state (s_t) at time t ,*

$$\inf_{Z \in \mathcal{Z}} \Pr[a \in \Omega(s_t) | h_t] \geq \beta.$$

Proof sketch.: By construction of the propagated envelope, $\mathcal{B}_t \subseteq \hat{\mathcal{B}}_t$ for all t . Therefore, for each perception uncertainty model $Z \in \mathcal{Z}$, the history h_t induces a posterior belief $b_t^Z \in \mathcal{B}_t \subseteq \hat{\mathcal{B}}_t$, and the probability that a is Ω -admissible under that posterior is exactly the linear score $\phi_a(b_t^Z)$. Hence the worst-case conditional admissibility probability over $Z \in \mathcal{Z}$ is lower bounded by the minimum of ϕ_a over the envelope, namely $\underline{p}_t(a)$, which is at least β whenever $a \in \hat{\Xi}(h_t)$. The full proof appears in the supplementary material. In words, any action admitted by the deployed shield is guaranteed to be Ω -admissible with probability at least β for every perception uncertainty model in the admissible set.

Conservativeness.: If $a \notin \hat{\Xi}(h_t)$, then either $a \notin \Xi^*(h_t)$, or a is excluded due to envelope/relaxation over-approximation.

Thm V.1 and *Conservativeness* hold for any perfect-perception shield Ω . To obtain an explicit finite-horizon safety lower bound, we instantiate Ω with a PCIS-style invariance condition.

Assumption V.1 (PCIS Instantiation: One-Step Bound). *We instantiate the perfect-perception shield Ω using the one-step PCIS-style admissibility constraint in Equation (2). Assume that there exists a designated safe subset $C \subseteq S$ and a constant $\gamma \in (0, 1]$ such that for all $s \in C$ and $a \in \Omega(s)$,*

$$\sum_{s' \in C} T(s, a, s') \geq \gamma.$$

Thus, whenever the current state lies in C and an action admitted by Ω is taken, the next state remains in C with probability at least γ .

Theorem V.2 (Finite-Horizon Double Probability). *Let D denote the random training dataset, and let $\hat{\Xi}^D$ and $\mathcal{Z}^D$ be the resulting deployed shield and admissible set of perception uncertainty models. Assume that the support of the initial belief b_0 is contained in C , actions are chosen from $\hat{\Xi}^D$, and Assumption V.1 holds. Then for horizon H ,*

$$\Pr_D[\Pr[\text{Safe}_{0:H}] \geq (\beta\gamma)^H] \geq \lambda.$$

In words, the outer probability is over the randomness in the training dataset used to construct the interval bounds. With probability at least λ over that dataset randomness, the resulting deployed shield guarantees closed-loop safety over horizon H with probability at least $(\beta\gamma)^H$. Here $\text{Safe}_{0:H}$ denotes the event that the trajectory starts in the safe subset C and remains in C through time H . (This is a finite-horizon guarantee, and the lower bound can decay quickly as H grows.)

Proof sketch.: Condition on the model-correctness event that the true perception uncertainty model lies in the admissible set of perception uncertainty models. Along any safe prefix, Abstraction Soundness guarantees that the deployed shield selects an Ω -admissible action with probability at least β , while Assumption V.1 gives probability at least γ of remaining in the safe subset C once such an action is taken. Therefore each step preserves safety with probability at least $\beta\gamma$, and a chain-rule argument over H steps yields the conditional lower bound $(\beta\gamma)^H$ on $\Pr[\text{Safe}_{0:H}]$. Finally, Section IV gives the outer probability bound λ for the

model-correctness event, producing the stated double-probability guarantee. The full proof appears in the supplementary material.

Thus the theorem should be read as a finite-horizon design certificate whose usefulness depends on the design point. Since the lower bound scales as $(\beta\gamma)^H$, the threshold β , the one-step invariance level γ , and the horizon H must be chosen carefully for the guarantee to remain meaningful; for lower per-step values or longer horizons, the formal lower bound can become vacuous even when empirical safety remains high. Section VI therefore compares the numerical certificate values with empirical rollout safety to show when the bound is informative.

State quantization.: The theorems above are stated for a finite IPOMDP. If the original system has continuous state or action variables, a separate abstraction step is needed before applying the shield. Existing stochastic-abstraction methods can provide probabilistic error bounds between a continuous stochastic system and a finite abstraction [32], [33]. Such bounds can in principle be composed with our interval-perception guarantee by treating the abstraction error as an additional dynamics-side error term in the construction of the safe set or in the one-step lower bound γ . Developing this composition for a particular continuous abstraction method is outside the scope of the present paper; here our guarantees apply to the finite models used by the shield.

VI. EXPERIMENTAL EVALUATION

A. Setup

We evaluate five shielding methods on four benchmark autonomous systems, usually modeled as MDP or POMDP, but here represented as IPOMDPs. These benchmarks are chosen to vary the ratio of states (S) to observations (O) and the scale of the underlying decision problem: TaxiNet (16 states, 16 observations), Obstacle (50 states, 3 observations), CartPole (82 states, 82 observations), and Refuel (344 states, 29 observations). TaxiNet is a runway-alignment benchmark in which safety means staying within the cross-track and heading-error bounds. Obstacle is a grid-navigation benchmark in which safety means avoiding obstacle cells. CartPole is a balance-control benchmark in which safety means remaining within the standard CartPole position and pole-angle limits. Refuel is a gridworld with fuel constraints in which safety means avoiding both obstacle collision and fuel exhaustion away from a refuel station.

Obstacle and Refuel are adapted from prior support-shielding case studies [8], but their observation kernels are modified here to fit the interval-perception setting and allow us to study the effect of shielding. In particular, the original Refuel benchmark could be solved nearly perfectly by an RL-trained controller, leaving little room for shielding to improve safety. We therefore modify Refuel to remove direct observation of the crash predicate and of whether fuel is nonzero, so the safety-critical variables are latent and must be inferred from history rather than read directly from the current observation. TaxiNet uses observation uncertainty derived from real autonomous-taxiing perception-confusion data [1][3], [2]. CartPole likewise uses a learned perception model, but in this case we train the perception model ourselves for the benchmark. Obstacle and Refuel instead use synthetic observation uncertainty, meaning that we begin from a hand-specified observation function and perturb it with an observation-noise budget to obtain interval-valued observation probabilities. These domains stress different

partial-observability regimes, from near-bijective observation kernels (CartPole) to severe compression of many states into a few observations (Obstacle) and latent safety-critical variables that must be inferred from history (Refuel). In each case, the dynamics are fixed and the observation kernel uses confidence intervals, yielding the IPOMDP used in evaluation; unless otherwise noted, these are Clopper–Pearson intervals at significance level $\alpha = 0.05$.

The continuous-state benchmarks are first converted to finite models before applying our shielding construction. TaxiNet uses the TaxiNet abstraction from prior work [1], [3], [2], [17]: cross-track error is discretized to $-2, \dots, 2$, heading error to $-1, 0, 1$, and an absorbing failure state gives 16 states. CartPole uses three uniform bins for each of position, velocity, pole angle, and angular velocity over the artifact ranges $[-0.318, 0.397]$, $[-2.288, 2.550]$, $[-0.258, 0.266]$, and $[-3.094, 3.290]$, giving $3^4 + 1 = 82$ total states. Obstacle and Refuel are finite gridworlds: Obstacle uses a 7×7 grid plus failure, and Refuel uses a 7×7 grid with fuel levels $0, \dots, 6$ plus failure. The induced PCIS one-step bounds are $\gamma = 1$ for all four benchmarks: the PCIS shields admit only actions whose dynamics support remains safe in the finite models used for evaluation.

We compare five shielding methods by measuring failure, stuck, and safe rates under the same RL-trained controller and perception regime; additional evaluation details appear in the supplementary material.

B. Baselines

The comparison includes two prior reference points and three methods introduced in this paper. The prior baselines are the *Observation* shield and the support-based shield of Carr *et al.* [8]. By contrast, *Single-Belief*, *Fwd-Sampling*, and *Envelope* are all novel methods in this work: *Single-Belief* is our point-estimate history-based shield, *Fwd-Sampling* is our sampled under-approximation to reachable interval beliefs, and *Envelope* is our LP-based over-approximation shield.

The *Observation* shield is memoryless: at each step it computes a posterior distribution over states using only the current observation, a uniform prior, and the point-estimate perception model, and it allows exactly those actions whose probability of being safe under that posterior exceeds the threshold. The point-estimate perception model is built from expected emission probabilities. The *Single-Belief* shield maintains a standard Bayesian belief under point-estimate observations and applies the same threshold test to the current belief. The proposed *Envelope* shield maintains a template polytope over reachable beliefs under interval uncertainty and admits an action only when its worst-case safety score exceeds the threshold. The *Fwd-Sampling* shield instead tracks an under-approximation of the reachable belief set using forward-sampled concrete belief points; it maintains a budget of $N = 500$ belief points and uses $K = 100$ sampled observation-probability vectors per propagation step. Finally, we include the support-based shield of Carr *et al.* [8], computed on the point-estimate POMDP whenever the support-MDP construction is feasible. That construction builds an MDP whose states are reachable belief supports and computes a winning region offline.

Envelope propagation is practical for TaxiNet and Obstacle, but not for CartPole and Refuel. Fwd-Sampling is feasible on all four benchmarks. Carr is feasible for TaxiNet, Obstacle, and

CartPole but infeasible on Refuel because the number of reachable supports in the support-MDP becomes too large during the offline breadth-first exploration, and on TaxiNet it degenerates because the point-estimate POMDP has no winning support.

This feasibility split shapes the interpretation. *Envelope* is the strictest and most expensive method because it certifies actions against an over-approximation of all reachable beliefs under interval uncertainty. We therefore use it as the strongest robust reference point when tractable, while *Fwd-Sampling* is the scalable approximation at much lower online cost. Evaluating both separates what stricter interval-belief reasoning buys from how closely cheaper sampling recovers it in practice.

C. Evaluation Protocol

To evaluate each shield, we run Monte Carlo rollouts of the closed-loop system consisting of the IPOMDP, an RL-trained controller, and the shield. In each rollout, the current history is passed to the controller, the shield filters the controller’s proposed action, the next latent state is sampled from the dynamics, and the next observation is sampled from the perception mechanism. Because the observation probabilities are interval-valued, we consider two ways of instantiating perception during evaluation. In the *uniform* regime, each rollout samples a valid observation kernel within the interval constraints and then samples observations from that kernel throughout the rollout. In the *adversarial* regime, we first solve an offline optimization problem over fixed interval realizations and then hold the resulting realization fixed for the entire rollout. Concretely, we optimize over observation kernels within the interval bounds using a cross-entropy method to maximize the empirical failure rate against the deployed shield-compliant RL controller. We do not use a time-varying adversary that chooses a new observation kernel at each step: such an adversary would match the worst-case variability already allowed by our interval-belief shields and would therefore make these interval-belief shields appear stronger relative to evaluations under a fixed-kernel perception model.

All reported numbers use an RL-trained controller from our experimental implementation. The controller is a Deep Q-Network trained directly on the IPOMDP dynamics with observation-action history as input, using a safety-oriented reward that penalizes reaching `FAIL`, rewards completing the finite horizon safely, and gives a small per-step reward for survival. The controller proposes an action and the shield either admits it or replaces it by a random admissible action; details appear in the supplementary material. We sweep $\beta \in \{0.50, 0.60, 0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 0.95\}$ for all threshold-based shields, run 200 Monte Carlo rollouts per condition, and report three mutually exclusive outcomes: *fail* (the trajectory reaches a `FAIL` state), *stuck* (the trajectory reaches a state where the shield blocks every action), and *safe* (neither event occurs within the benchmark horizon). The benchmark horizons are 20 steps for TaxiNet, 25 for Obstacle, 15 for CartPole, and 30 for Refuel.

For threshold-based shields we report the operating point that minimizes fail rate, with stuck rate as a tiebreaker, and we use the per-threshold sweeps to discuss the corresponding stuck-avoidance tradeoffs.

D. Cross-Case Picture

Figure 1 summarizes the lowest-failure setting across the four benchmarks. A key driver of the observed differences appears to be observation informativeness: how strongly the current observation narrows down the latent state. When observations almost identify the latent state, as in CartPole, all feasible shields are nearly equivalent and incur essentially no stuck-rate penalty. At the opposite extreme, Obstacle compresses 50 states into only 3 observations; memoryless methods then lose any useful middle ground between allowing risky actions and blocking almost everything, while belief history appears necessary. TaxiNet sits between these extremes: it has as many observation labels as states, but the learned perception model is noisy enough that one-step posteriors remain ambiguous. Refuel is different again: the observation kernel hides the variables that determine whether collision or fuel exhaustion is imminent, so history is useful but even accurate belief tracking can lead to paralysis.

The proposed envelope construction improves the safety side of the tradeoff whenever it is feasible to run. Here each shield is evaluated at its own threshold selected by the low-failure rule of Fig. 1, so the comparison is between each method’s preferred safety-first operating point rather than at a common shared threshold. On TaxiNet, Envelope remains the strongest robust option, reaching 25.5% fail and 21.5% stuck under uniform perception, compared with 35.5% fail and 15.5% stuck for Single-Belief and 26.5% fail and 26.5% stuck for Fwd-Sampling; under adversarial perception the same ordering remains, at 27.0%/34.5% for Envelope versus 37.5%/14.0% for Single-Belief and 34.5%/22.0% for Fwd-Sampling. On Obstacle, the gain is larger: Envelope reaches 6% fail and 80% stuck under uniform perception, while Single-Belief remains at 16% fail and 52% stuck and Fwd-Sampling improves to 8% fail and 86% stuck. Under adversarial perception, Envelope reaches 8% fail and 78% stuck, still clearly improving on Single-Belief’s 12%/56%. This extra stuck behavior is the empirical counterpart of the over-approximation soundness result: because the envelope certifies actions against all beliefs in an over-approximation of the reachable set, it can block actions that look safe under favorable point-estimate beliefs but are not safe in the worst case.

The adversarial columns show that worst-case observation realizations affect benchmarks unevenly. They are harsher on history-based shields in TaxiNet and, less so, Obstacle, but remain closer to uniform results in CartPole and Refuel. This pattern is consistent with TaxiNet and Obstacle leaving more room for an optimized observation kernel to interfere with the closed-loop policy.

Figure 2 provides the complementary stuck-avoidance view. Once the threshold is selected to maximize safe completions rather than minimize failures, the ranking changes in the direction suggested by the low-failure figure. On TaxiNet and Obstacle, Single-Belief or Fwd-Sampling usually become preferable to Envelope, suggesting that the extra conservatism of Envelope can convert many potentially safe runs into stuck runs. On Refuel, the strongest stuck-avoidance picture is dominated by Observation, which reaches about 97% safe under uniform perception and 98% under adversarial perception at low thresholds, whereas the zero-failure operating points of Single-Belief and Fwd-Sampling have much higher stuck rates. CartPole remains near-degenerate across all shields, with every method achieving roughly 97–99% safe completions and no stuck

episodes at its best threshold.

The remaining baselines illustrate complementary failure modes. The Observation shield can be competitive when a single observation already localizes the state, but it degrades sharply under aliasing, meaning that multiple latent states produce the same or very similar observations. In TaxiNet, its low-failure point (8.5% fail under uniform perception) is only obtained at 91.5% stuck, which is still a much less attractive tradeoff than what the envelope achieves by using history quantitatively. In Obstacle, Observation becomes almost indistinguishable from support-only shielding, both ending at approximately 1–2% fail and 98–99% stuck. In Refuel, however, Observation has a different role: at lower thresholds it is still the only shield that attains a 0%-stuck operating point, albeit at 3–4.5% failure. Fwd-Sampling sits between Single-Belief and Envelope on TaxiNet and Obstacle, but on Refuel it behaves much more conservatively, with 0% fail at the cost of 100% stuck under uniform perception and about 89% stuck under adversarial perception. Carr’s shield is strongest when support information is already nearly enough to identify the state, but it can collapse entirely once every reachable support contains conflicting safety requirements. TaxiNet is the clearest example: under Carr’s offline support-MDP analysis on the point-estimate POMDP, the winning region is empty, so Carr effectively blocks almost all runs from the initial step.

E. Comparison with Support-Based Shielding

The comparison with Carr-style support shielding clarifies when probability mass is important. Support reasoning can be highly effective when observations are informative enough that reachable supports stay small; the CartPole results are consistent with this, since Carr remains competitive there. But support-only reasoning discards the distinction between a state with posterior weight 0.49 and one with posterior weight 10^{-4} , and in aliased problems that distinction can determine whether a useful action should still be allowed. TaxiNet makes this failure mode stark: support shielding has no winning support at all, even though the envelope, Fwd-Sampling, and Single-Belief filters all retain nontrivial safe behavior. Obstacle shows the softer version of the same phenomenon, where support reasoning remains feasible but collapses to near-total stuck.

Taken together, these experiments support a narrow but important claim. Interval belief envelopes are not a universal replacement for simpler shields. Instead, they appear most useful in the regime where point-estimate belief filters are too optimistic and support abstractions are too coarse. CartPole suggests that they are unnecessary when observations already localize the state well. TaxiNet and Obstacle illustrate where they can be useful. Refuel shows the computational boundary where heavier robust shielding becomes impractical. Fwd-Sampling is a useful intermediate point in that design space: it is far cheaper than the LP envelope and available on all four benchmarks, but because it under-approximates the reachable belief set it generally trades extra blocked actions for only limited safety improvement. In the difficult aliased regimes studied here, the envelope provides the strongest safety performance among the feasible shields, and its extra conservatism can be quantified rather than treated as a black box.

F. Conformal Shielding Comparison

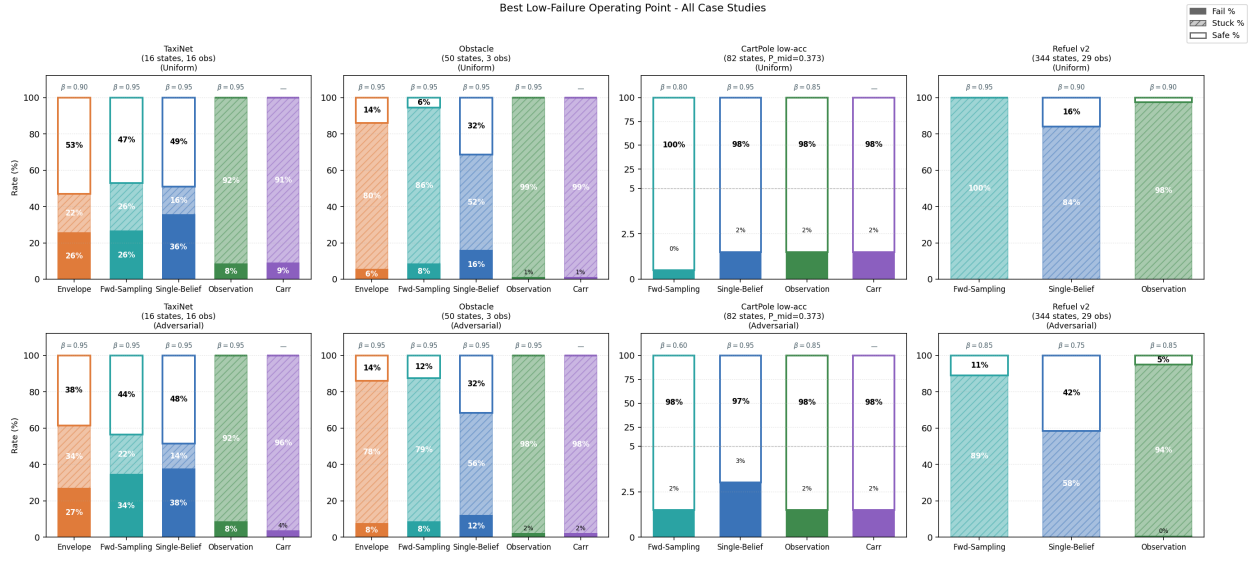


Fig. 1. Best low-failure operating point of each shield under uniform and adversarial perception. The selected threshold minimizes fail rate and then stuck rate. Observation informativeness determines which shield family is useful, and adversarial harshness varies by case study. Only CartPole uses a disjoint piecewise vertical scale, expanding 0–5% and compressing 5–100%; other case studies use the standard scale. The corresponding numerical values and uncertainty intervals appear in supplementary Table S-I.

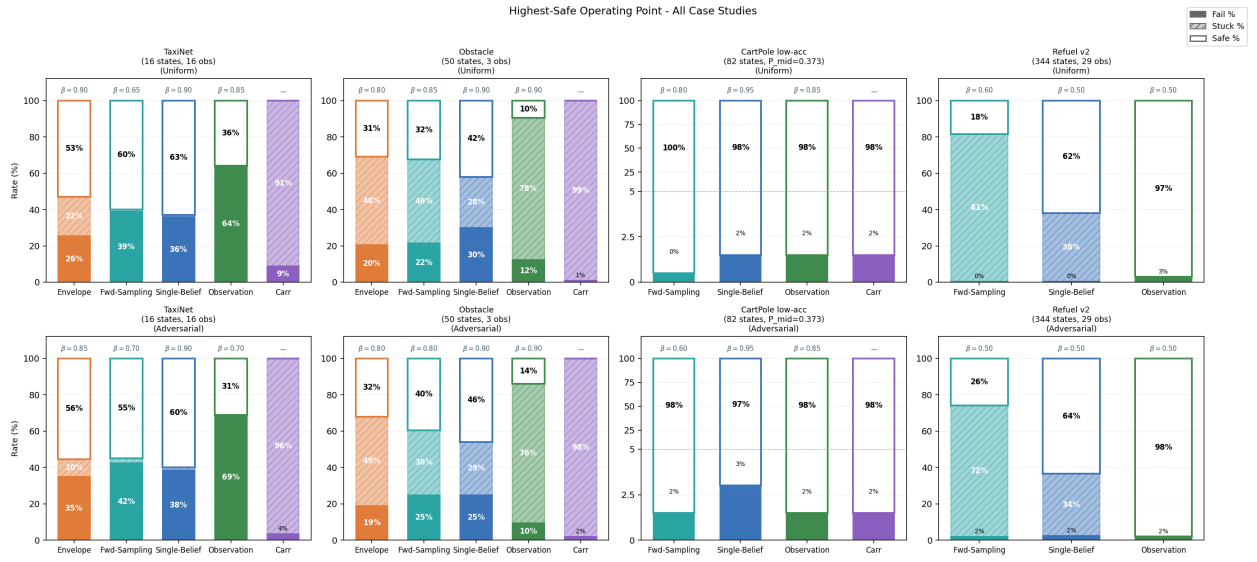


Fig. 2. Highest-safe operating point of each shield. The selected threshold maximizes safe completions, with lower fail and then lower stuck as tiebreakers. Solid color denotes fail, hatching denotes stuck, and the outlined top segment denotes safe. Threshold labels match Fig. 1. Only CartPole uses the disjoint piecewise vertical scale; other case studies use the standard scale. The corresponding numerical values and uncertainty intervals appear in supplementary Table S-II.

We also compare the interval-belief shields with the TaxiNet conformal shielding baseline of [12]. The comparison keeps the same finite TaxiNet IPOMDP dynamics, action space, controller, and rollout protocol, but substitutes the simulated perception artifact from [12]. That artifact provides paired samples $(s, \hat{s}, C_\kappa)$, where s is the true discretized state, $\hat{s}$ is the point estimate consumed by our shields, and C_κ is the conformal prediction set consumed by the conformal shield. We model this as $P(\hat{s}, C_\kappa | s) = Z(\hat{s} | s) Q_\kappa(C_\kappa | s, \hat{s})$: the interval-belief shields use Z , while the conformal shield samples its set conditionally on the realized $(s, \hat{s})$, including under adversarial point-estimate realizations. This preserves the empirical dependence between point estimates and conformal sets; the supplementary material gives the full construction.

Figure 4 suggests a distinct tradeoff for the conformal baseline. At the lowest-failure operating point, the high-confidence conformal shield has lower failure rates than the interval-belief shields, but most rollouts become stuck: 12.5% fail and 85.5% stuck under uniform perception, and 15.5% fail and 84% stuck under adversarial perception. The strongest interval-belief shield in this row is Envelope, with 32%/46% fail/stuck under uniform perception and 39%/41% under adversarial perception. At the highest-safe operating point, the conformal shield does outperform Envelope and Fwd-Sampling on safe completions, reaching 25% and 25.5% in the two regimes. However, this operating point still leaves about three quarters of rollouts failing, while Single-Belief reaches 38% and 41% safe completions. In this experiment, the conformal baseline

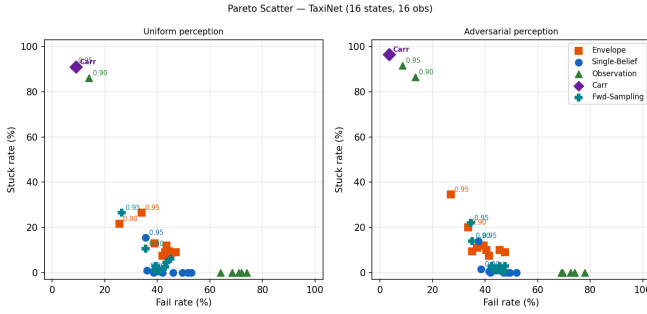


Fig. 3. TaxiNet Pareto scatter over threshold β under adversarial perception. History-based shields improve fail-versus-stuck tradeoffs over memoryless Observation, and Envelope improves on Single-Belief by removing actions safe only for the point-estimate model.

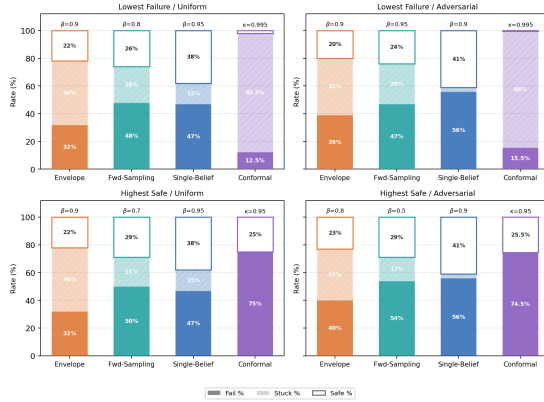


Fig. 4. TaxiNet comparison under uniform and adversarial perception. The top row selects each method’s lowest-failure operating point, while the bottom row selects each method’s highest-safe operating point. The interval-belief shields (Single-Belief, Envelope, Fwd-Sampling) are swept over threshold β ; the Conformal baseline is swept over confidence level κ . The setting that each method selects in each panel is annotated above its bar. The corresponding numerical values and uncertainty intervals appear in supplementary Table S-III.

appears to reduce failures mainly by blocking most rollouts at high confidence, while lower confidence improves liveness at the cost of a high failure rate. The interval-belief shields exhibit a different fail-versus-stuck tradeoff, consistent with their use of history and probability mass within the uncertainty model.

This experiment is empirical and does not test the exchangeability assumption behind conformal coverage under closed-loop policy changes; it compares the two approaches assuming no distribution shift. The high stuck rates for conformal shielding at high confidence are consistent with a method that does not account for observation history or state probabilities. In other words, conformal shielding appears to be a coarser method that preserves less information about the state distribution and therefore makes more conservative shielding choices.

G. Additional Analyses

Table I compares the finite-horizon certificate with empirical rollout outcomes for the tractable Envelope cases. The comparison should be read as a lower-bound sanity check, not as a prediction of rollout rates: the theorem certifies remaining in the safe state set, whereas the plotted safe-completion outcome excludes both failures and stuck rollouts. For that reason the table compares the

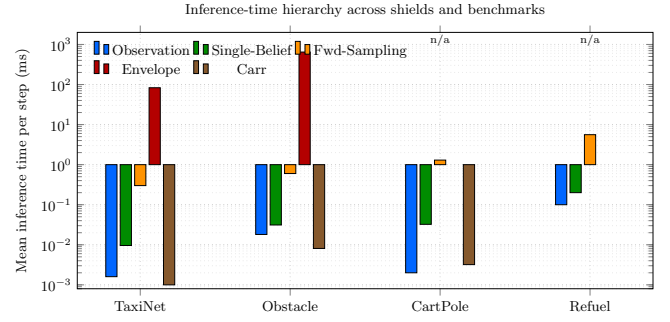


Fig. 5. Mean shield inference time per step on a log scale. Observation, Single-Belief, and Carr remain in the microsecond regime; Fwd-Sampling moves into the low-millisecond regime; and the LP-based Envelope is substantially slower, with CartPole and Refuel beyond the feasible range.

certificate with empirical non-failure rather than with the safe-completion segment of the stacked bars. The certificate is well below the observed non-failure rates across the tested thresholds, as expected, because the proof compounds a worst-case one-step lower bound over the full horizon.

The supplementary material reports two further diagnostics. First, a TaxiNet sweep over the Clopper–Pearson perception-interval significance level $\alpha \in \{0.01, 0.025, 0.05, 0.075, 0.10, 0.15, 0.20, 0.30\}$ shows that β is the dominant tuning parameter and that α mainly fine-tunes interval conservatism within the operating regime selected by β . Second, an envelope-conservatism study compares the LFP envelope with forward-sampled under-approximations under fixed-kernel and time-varying perception semantics, separating the conservatism due to belief-set over-approximation from the additional pessimism of protecting against time-varying Z_t .

H. Inference-Time Tradeoffs

The fail-versus-stuck tradeoffs in Figs. 1 and 2 are mirrored by the runtime hierarchy in Fig. 5. These timings measure one shield inference call, including belief propagation and action filtering but excluding controller inference, environment stepping, and observation sampling, on an AMD EPYC 9454P. Observation, Single-Belief, and Carr are in the microsecond regime; Fwd-Sampling is in the millisecond-to-subsecond regime; and LP-based Envelope is two to five orders slower where feasible. On TaxiNet, for example, mean latency is $9.4\mu\text{s}$ for Single-Belief, 21.2 ms for Fwd-Sampling, and 83.1 ms for Envelope. Refuel is the scalability boundary: Observation and Single-Belief remain sub-millisecond, Fwd-Sampling reaches 383.9 ms , and Envelope and Carr are infeasible.

These timings sharpen the engineering interpretation. Single-Belief appears to be the strongest lightweight default in these benchmarks: nearly as cheap as Observation while retaining history and generally outperforming memoryless shielding. Fwd-Sampling is a useful middle ground on smaller models, much faster than Envelope on TaxiNet and Obstacle while recovering much of its safety benefit, but its cost grows steeply with state dimension and interval coverage. Envelope is best viewed as a low-frequency or smaller-model mechanism for hard aliased regimes, while Fwd-Sampling is the scalable approximation when exact envelope propagation is unavailable but more conservatism than Single-Belief is desired.

TABLE I

FINITE-HORIZON CERTIFICATE VALUES COMPARED WITH EMPIRICAL SAFETY FOR THE ENVELOPE SHIELD AT THREE RUNTIME THRESHOLDS. UNCERTAINTY IS ONE BINOMIAL STANDARD DEVIATION OVER 200 ROLLOUTS. EMPIRICAL NON-FAILURE, 1—FAIL, IS THE CLOSEST ANALOGUE OF $\text{Safe}_{0:H}$, BECAUSE STUCK ROLLOUTS ARE LIVENESS FAILURES BUT NEED NOT LEAVE THE SAFE STATE SET. SINCE THE INDUCED PCIS ONE-STEP BOUND IS $\gamma=1$ FOR THESE BENCHMARK MODELS, THE CERTIFICATE COLUMN REPORTS β^H .

Case study	Regime	β	H	Fail	Stuck	Emp. non-failure	β^H	Gap
TaxiNet	uniform	0.85	20	0.435 ± 0.035	0.120 ± 0.023	0.565 ± 0.035	0.039	0.526 ± 0.035
TaxiNet	uniform	0.90	20	0.255 ± 0.031	0.215 ± 0.029	0.745 ± 0.031	0.122	0.623 ± 0.031
TaxiNet	uniform	0.95	20	0.340 ± 0.033	0.265 ± 0.031	0.660 ± 0.033	0.358	0.302 ± 0.033
TaxiNet	adversarial	0.85	20	0.350 ± 0.034	0.095 ± 0.021	0.650 ± 0.034	0.039	0.611 ± 0.034
TaxiNet	adversarial	0.90	20	0.335 ± 0.033	0.200 ± 0.028	0.665 ± 0.033	0.122	0.543 ± 0.033
TaxiNet	adversarial	0.95	20	0.270 ± 0.031	0.345 ± 0.034	0.730 ± 0.031	0.358	0.372 ± 0.031
Obstacle	uniform	0.85	25	0.13 ± 0.023	0.65 ± 0.034	0.88 ± 0.023	0.017	0.858 ± 0.023
Obstacle	uniform	0.90	25	0.12 ± 0.023	0.71 ± 0.032	0.88 ± 0.023	0.072	0.808 ± 0.023
Obstacle	uniform	0.95	25	0.06 ± 0.016	0.81 ± 0.028	0.94 ± 0.016	0.277	0.668 ± 0.016
Obstacle	adversarial	0.85	25	0.19 ± 0.027	0.60 ± 0.035	0.82 ± 0.027	0.017	0.798 ± 0.027
Obstacle	adversarial	0.90	25	0.11 ± 0.022	0.71 ± 0.032	0.90 ± 0.022	0.072	0.823 ± 0.022
Obstacle	adversarial	0.95	25	0.08 ± 0.019	0.79 ± 0.029	0.92 ± 0.019	0.277	0.648 ± 0.019

VII. RELATED WORK

The closest related line of work is shielding under partial observability. Carr *et al.* [8] compute the support-based shield over belief supports, retaining only which latent states remain possible after an observation history and discarding emission probabilities. That support-based shield yields strong soundness guarantees for avoid and reach-avoid objectives and is an important baseline for our setting. Our contribution differs in the semantic object being propagated: instead of reachable supports, we propagate quantitative belief envelopes induced by interval-valued emission probabilities. This retains information that the support-based shield intentionally throws away, which is useful when support winning regions collapse or become too coarse to admit a usable shield.

Related work also abstracts learned perception by a compact probabilistic interface between latent state and downstream symbolic reasoning. Păsăreanu *et al.* [17] replace a vision stack by a confusion-matrix-derived stochastic abstraction to enable closed-loop probabilistic analysis, while Cleaveland *et al.* [26] construct conservative IMDP perception abstractions from finite data using confidence intervals and intra-bin probability enlargement for probabilistic model checking. Schäfers *et al.* [18] use learned perception uncertainty models to drive POMDP belief updates from visual observations. Peper *et al.* [34] likewise pursue PAC-style finite-sample guarantees for vision-based autonomy, unifying probabilistic verification and validation of learned perception components. Our work adopts the same modeling move of summarizing perception by state-conditional emission probabilities, but differs in both objective and semantics: rather than verifying an offline closed-loop stochastic model, we propagate interval-valued belief envelopes online and use them to enforce conservative shielding under partial observability.

Adjacent work on predictive uncertainty under shift clarifies why we use interval bounds on the perception uncertainty model rather than conformal wrappers. Conformal prediction gives finite-sample marginal coverage for labels or outputs under exchangeability [11], [10], with extensions for covariate shift [13], [35], online drift [14], hidden Markov models, and off-policy MDP evaluation [36], [37]. Thus conformal methods are not ruled out merely by control-based shift. However, their certified objects are prediction sets, predictive distributions, or policy-value intervals, not entry-wise bounds on the latent-state perception model $Z(o|s)$ needed for belief propagation under deployment shift. Work using conformal wrappers for action

shielding [12] could be improved for control-based shift, but would still struggle when deployment changes the conditional emission probabilities themselves. Our method isolates uncertainty at that conditional level, tying the guarantee to probable realizations of per-state emission probabilities rather than the aggregate deployment state distribution. Ovadia *et al.* [15] further motivate this caution by showing that point-valued uncertainty estimates can degrade under shift.

More broadly, our work sits in the shielding literature [4], [5], [6] and uses standard POMDP belief semantics [16], [38]. It combines robust/imprecise probabilistic modeling [19], [20], [21], Clopper–Pearson finite-sample intervals [22], and state-based shields such as PCIS [23] or barrier-style invariance methods [24] into a runtime imperfect-perception shield built by lifting a perfect-perception shield through interval-valued belief envelopes.

VIII. CONCLUSION

We presented a shielding method for imperfect-perception agents modeled as IPOMDPs where the perception uncertainty model is learned from finite data and represented by confidence intervals rather than point estimates. The method lifts a perfect-perception shield into belief space by propagating conservative belief envelopes and admitting only actions whose worst-case shield score remains above threshold. For the finite discrete models considered here, this gives a runtime realization of a fundamentally history-based shield that scales much better than explicit history reasoning. It yields a finite-sample outer guarantee on model correctness together with a conditional lower bound on runtime safety, while preserving a modular interface to upstream state-based shield constructions such as PCIS. Detailed proofs, expanded benchmark and protocol descriptions, the complementary stuck-avoidance summary, and additional diagnostic evaluation appear in the supplementary material.

The empirical picture is that interval belief envelopes occupy a useful middle ground between point-estimate methods and the support-based shield. Relative to single-belief baselines, they provide more reliable action correction under perception uncertainty; relative to the support-based shield, they remain usable in regimes where support abstractions collapse or become too coarse. At the same time, the experiments make the price of this additional fidelity visible: Envelope propagation is computationally demanding, infeasible on the largest benchmark settings, and sensitive to the

coarseness of the template abstraction used to represent reachable beliefs. Fwd-Sampling is a scalable alternative that mimics some of the conservatism of LP envelope shielding when Envelope propagation is unavailable, but it is an under-approximation and does not provide the sound over-approximation guarantee. The benchmark comparison sharpens where this extra machinery matters: CartPole shows that it is unnecessary when observations already localize the latent state well, TaxiNet and Obstacle show the partially observable regimes where sound envelope propagation improves the safety-usability tradeoff, and Refuel illustrates the scalability limit of the strongest robust method.

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